

Received 27 March 2025, accepted 21 April 2025, date of publication 14 May 2025, date of current version 27 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3570167

RESEARCH ARTICLE

Complex Emotion Estimation Using Analysis-by-Synthesis of Facial Expression Images

WIN SHWE SIN KHINE¹, PRARINYA SIRITANAWAN², (Member, IEEE),
AND KAZUNORI KOTANI³

¹School of Information Science, Japan Advanced Institute of Science and Technology, Nomi 923-1211, Japan

²Graduate School of Science and Technology, Department of Engineering, Informatics and Interdisciplinary Systems Division, Shinshu University, Nagano 380-8553, Japan

³Faculty of Transdisciplinary Science, Institute of Philosophy in Interdisciplinary Science, Kanazawa University, Kanazawa, Ishikawa 920-1192, Japan

Corresponding author: Prarinya Siritanawan (prarinya@shinshu-u.ac.jp)

This work was supported by JSPS KAKENHI under Grant 23K16925.

ABSTRACT This research proposes an approach for recognizing facial expressions for complex emotion estimation through the utilization of generated facial images. The proposed approach consists of two main parts: facial image generation and facial expression recognition. In the first part, we introduce Conditioned Emotion Generative Adversarial Networks (cEmoGANs) to synthesize images that convey complex facial expressions. Unlike previous methods, our generative model maintains face identity information and expresses various types of complex emotions. This capability encourages the generator to have control over the image generation process, resulting in enhanced image quality, reduced distortion, and increased diversity of generated images. The second part involves the design of a multiple-label classification based on a convolutional neural network trained on the complex facial expression images generated from the first part, which is employed in the recognition of complex facial expressions for emotion estimation. Our model, using images generated by cEmoGANs, demonstrates a notable performance surpassing the capabilities of the previous models in comparative evaluations.

INDEX TERMS Complex emotions, analysis-by-synthesis, conditioned emotion generative adversarial networks, emotions estimation.

I. INTRODUCTION

Emotions are complex psychological states that have a significant impact on human-to-human communication. Effective interpersonal communication involves not only exchanging spoken words but also understanding the emotions expressed by the other party. In some instances, the emotions underlying a conversation hold more significance than the words themselves. Since emotions greatly impact social interactions, the ability to understand them is essential for effective communication.

People naturally learn to understand others' emotions while growing up in society. With the advancement of modern

machine learning and signal processing technology, machines can also learn to recognize emotions through various signals associated with these emotions. These emotional signals can be manifested in several forms, including voice, vocal tone, spoken words, heartbeat, blood pressure, facial expressions. Previous research has examined the significance of these different signals including the key study by Mehrabian and Ferris [1] reported that 55% of the conveyed signals come from facial expressions.

Although it is evident that facial expressions are important social signals in perceiving emotions, some argue that they are a conscious social practice instead of reflecting inner emotions. For example, a gentle smile can be used as a response to greet friends or colleagues in some cultures instead of reflecting the emotion of happiness. However,

The associate editor coordinating the review of this manuscript and approving it for publication was Szidónia Lefkovits¹.

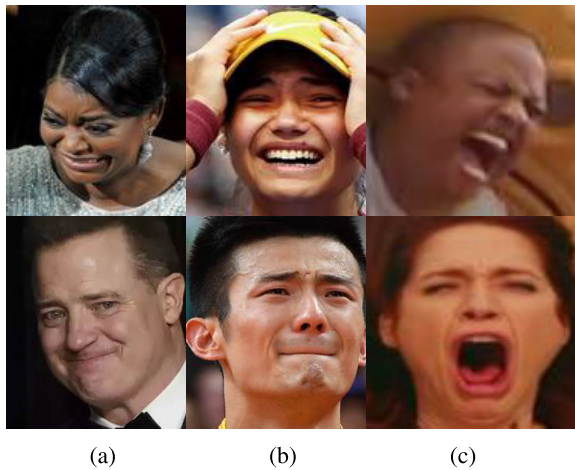


FIGURE 1. Facial expressions of mixed emotions (happiness surprise, happiness sadness and fear happiness) during event such as (a) the Oscars (b) the Olympic games and (c) riding a roller coaster [55], [56], [57], [58], [59], [60].

research [2] on visually impaired athletes has shown that they express the same facial expressions as others for winning and defeating games. Their results concluded that the occurrence of facial expressions does indeed reflect the underlying emotions.

The representation of emotions can be organized into dimensional and categorical models. In the dimensional model, emotions are represented using dimensional parameters. For instance, in Russell's work [3], emotions are represented by valence and arousal as the parameters. In the literature, the number of dimensions can vary, with some studies suggesting up to five dimensions [4].

In the categorical model, each emotion is described as an individual category. For example, psychologist Paul Ekman introduced anger, disgust, fear, happiness, sadness, and surprise as basic emotions that can be observed in human beings. The number of basic emotions can vary based on different beliefs about emotions. Among them, Ekman's six basic emotions are widely used in the recognition of facial expressions for emotion estimation as they are proven to be universal in [5].

Although Ekman's emotions are simple and basic, they cannot explain all emotional experiences encountered by human beings. For example, when we come across an unexpected joyful moment, we feel not only happiness but also sadness (see Fig. 1(a)). The images in Fig. 1 capture the facial expressions under emotional events, such as an actor receiving an Oscar award, an athlete winning a gold medal at the Olympic games, or someone riding a roller-coaster. These kinds of emotions cannot be solely defined by a single basic emotion category and they should be referred to as *complex emotions*. More types of complex emotions can be discovered in [6] and [7].

While research about basic emotions has been extensively studied in the past, most of the emotion recognition systems

commonly found in the literature can estimate the six basic emotions very well under constrained environments. However, the recognition of complex emotions has a challenging issue for such systems. In this research, we aim to develop an approach for estimating complex emotions using an analysis-and-synthesis approach, which involves two parts. The first part involves generating facial images that represent the facial expressions for complex emotions, while the second part focuses on the estimation of complex emotions by learning from the generated facial images. In contrast to prior research on complex emotions, the inputs of the generation model can be used to control both the strength and types of facial expressions, resulting in a more flexible model. We later demonstrate that images generated by the proposed generation model help to improve the performance of complex emotions estimation compared to using images generated by other models.

II. RELATED WORK

This section provides an overview of fundamental deep learning networks used in facial expression recognition. One of the earliest deep learning methods in this domain is the Deep Belief Networks (DBNs), proposed by Hinton et al. [8]. The method was developed to address the issues of slow learning rates in deep networks during back-propagation. DBNs, which consist of Restricted Boltzmann Machines (RBMs), are unsupervised graphical models designed to capture complex patterns in training data. Each layer in a DBN learns representations of the entire input, making them applicable for image recognition tasks. The method was adopted for facial expression recognition in the earlier deep learning-based implementation [9], [10], [11].

As deep learning research evolved, Convolutional Neural Networks (CNNs), introduced by LeCun et al. [12], emerged as a dominant approach for image recognition tasks involving large scale real-world image samples. CNNs utilize convolution filters with properties like sparse interaction, translation equivalence, and parameter sharing, along with down-sampling operations and fully connected layers, resulting in a high performing recognition system. These capabilities make CNNs suitable for a wide range of applications, from general image classification [13] to specialized tasks like visual inspection in industry [14]. Architectural variations of CNNs specifically designed for facial expression recognition can be found in [15], [16], [17]. Furthermore, CNNs have also demonstrated capabilities beyond processing image signals, extending to behavioral and emotional recognition tasks involving other physiological signals such as brain impulses captured via Electroencephalography (EEG) devices [18], heart rate variability [19], or audio signals from speech [20]. Alternatively, some of previous works have considered sequential data for facial expression recognition. To handle such data by deep learning methods, Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM)

networks, are commonly employed. RNNs are designed with internal memory that retains information from previous time steps, making them particularly effective for tasks involving time-series data, such as the analysis of human behaviors in video, including applications like sign language recognition [21]. LSTM, an extension of RNNs, addresses issues like exploding or vanishing gradients during training by incorporating gates regulating the cell state [22]. LSTM has been widely employed with the CNN based feature extraction for temporal facial expression recognition tasks from video data [23], [24].

Another widely used deep learning approach for facial expression recognition is the Deep Auto Encoder (DAE), an unsupervised learning model comprising symmetrical networks known as the *encoder* and *decoder*, which can serve as a feature extractor and generative model, respectively. DAE encodes data in a feature vector to reduce dimensions and learn significant representations. It is applied in tasks like noise reduction, dimensionality reduction, and facial expression recognition [25], [26]. Moreover, the decoder can reconstruct the feature vector back into the original image or even transform it into an image from a different domain. These characteristics are valuable for many applications such as semantic segmentation in autonomous driving [27], or medical diagnostics [28]. As the feature representations extracted by the encoder of DAE are highly informative and capable of capturing crucial information about an image in a compact form, these representations are often combined with image generation tasks, such as those facilitated by Generative Adversarial Networks (GANs) [40], [41], another key advancement in deep learning. The synergy between DAEs and GANs has been demonstrated in studies like [29], employing the discriminator's learned representations for reconstruction.

III. COMPLEX EMOTION ESTIMATION

In this section, we introduce an approach designed to make a facial expression recognition system that can estimate complex emotions using an analysis-by-synthesis approach. The approach consists of two primary parts: *facial image generation* for synthesizing complex facial expressions and *facial expression recognition*. An overview of the proposed approach is illustrated in Fig. 2. The first part is responsible for synthesizing complex facial expressions with additional control over face identity information and types of emotions, referred to as conditioned Emotion Generative Adversarial Networks (cEmoGANs). The second part focuses on recognizing facial expressions from the facial images generated by cEmoGAN to estimate complex emotions. Each part is subsequently explained in the following sections.

A. FACIAL IMAGES GENERATION

The first part introduces conditioned Emotion Generative Adversarial Networks (cEmoGANs) with the purpose of synthesizing facial images of complex emotions. In typical

image synthesis using GANs, image features are usually entangled within the latent space, meaning it is unclear which parts belong to facial expressions or facial features. To address this issue, we separate the features for facial expressions and face identity information using encoder and multiple emotion labels. This separation improves the quality of the generated images, as shown in the experimental section. In cEmoGANs, an encoder is used to transform a facial image into lower dimensional face identity information, while a six-dimensional vector is employed to control the categories of facial expressions.

The overview architecture of cEmoGANs is depicted in Fig. 3. It comprises three main components: an encoder, a generator, and a multidimensional label vector. The encoder (E) network configuration is illustrated in Fig. 4. It consists of two convolution layers. A linear activation is employed to ensure that its output follows a Gaussian distribution. The encoder E is trained to minimize the mean absolute error between the initially estimated latent sample and the one predicted by E .

Since the primary goal of E is to encode face identity information from the given image, the quality of the encoder depends on how close the bottleneck representation is to the original latent representation. Therefore, two metric are used to evaluate the encoder. The mean squared error (MSE) is used to measure the loss between the latents estimated by the encoder and the original latents. Additionally, it is important to assess the visual similarity of the generated image, which is based on the estimated latents compared to the original image given to the encoder. To quantify this, a cosine similarity score (CSS) is calculated to determine the facial similarity of generated images from estimated latents and given latents.

Hyperparameters such as number of layers, choice of optimizer and activation function, and batch size are empirically selected based on the results in Table.1. Based on those results, the encoder performs best with the proposed configuration as shown in Fig.4 and when trained with the Adam optimizer using a batch size of 64.

For the generator, unlike regular GANs [40], which generate images unconditionally and resemble random generation, the proposed cEmoGANs method use multidimensional conditional labels to control categories of emotional expressions in image generation. The generator G is based on the conditional GANs (cGANs) [30], which takes the condition label $\mathbf{y}$ and latent sample $\mathbf{z}$ for image generation as depicted in Fig. 5. cGANs is trained using Eq. (1) which uses label vectors $\mathbf{y}$ as conditions to control what to generate by G .

$$\mathcal{L}_{cGANs} = \mathbb{E}_{\mathbf{x} \sim p_{\mathbf{x}}} [\log D(\mathbf{x}|\mathbf{y})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}} [\log D(G(\mathbf{z}|\mathbf{y}))] \quad (1)$$

where $\mathbf{x}$ represents the ground truth images, and $\mathbf{y}$ represents the multidimensional labels provided as conditions. Adam optimizer was used to update the weights parameters of the network. In preliminary experiments, it was observed that the discriminator (D) tends to learn faster than the generator (G). As a result, a lower learning rate of 10^{-5} is set for D , while G uses a learning rate of 10^{-4} .

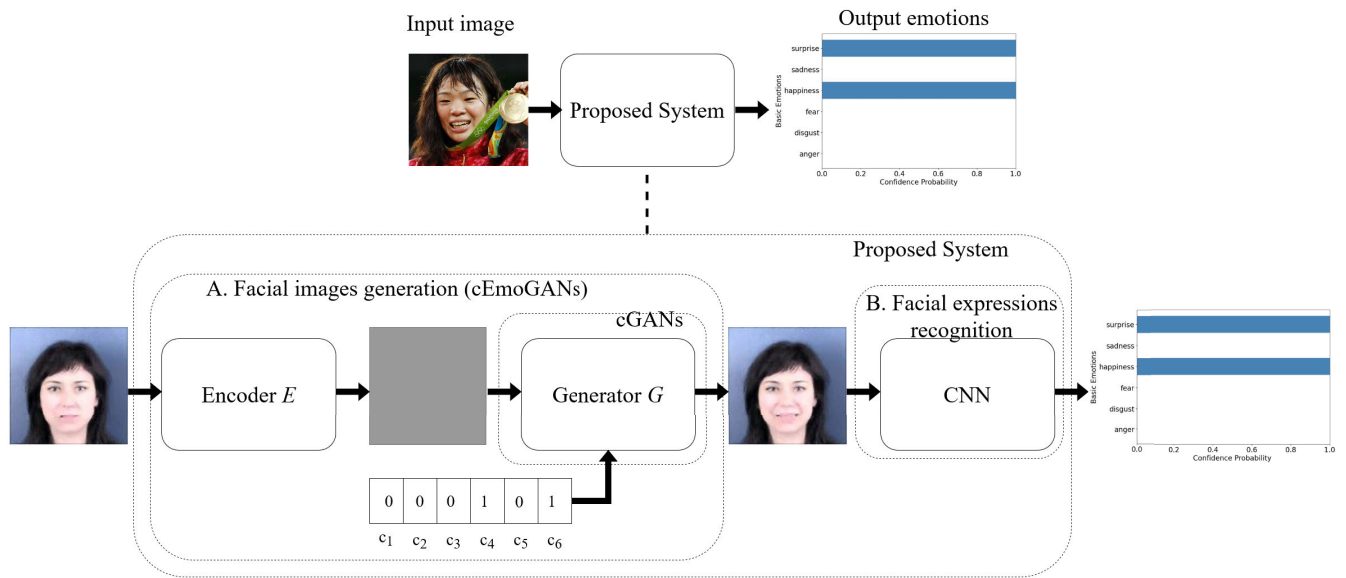


FIGURE 2. Overview of the proposed approach for complex emotion estimation.

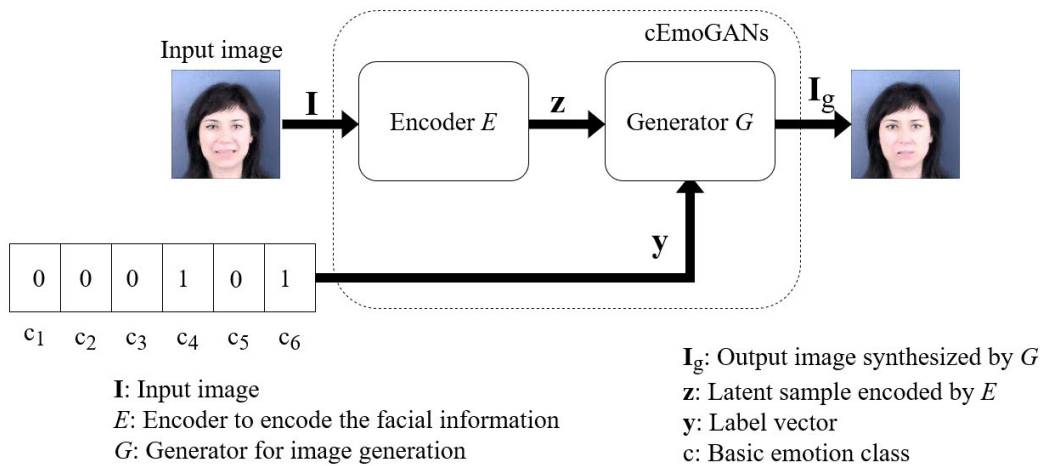


FIGURE 3. Overview of Conditional emotion generative adversarial networks (cEmoGANs).

During the training process, the generator G takes a latent sample $\mathbf{z}$, along with a six-dimensional label vector representing complex emotions. Each dimension corresponds to one of the basic emotions: anger, disgust, fear, happiness, sadness, and surprise, in that order. If a particular emotion is present in the input image, the respective dimension is set to 1; otherwise, it is set to 0. The discriminator D also takes this labels vector along with the input images. After training cEmoGANs, the trained generator G is used to synthesize facial images of complex emotions, which are later employed to train the facial expression recognition system in the next part.

As complex facial expressions in this research are yielded from a composition of different basic emotion categories, the image features of these mixed emotions encapsulate

the essential visual characteristics of each contributing emotion. Building on this concept, we synthesize complex facial expressions by ensuring that the facial features of both emotions are conditionally represented during image generation, enforced through conditional control.

While the concept of cGANs is adopted from [30], a key difference and the primary contribution of cEmoGANs is the encoder-decoder architecture, which enables the formulation of latent samples that maintain identity information during expression synthesis. Unlike standard cGANs, which can only generate images from single emotion categories and struggle with blended expressions, cEmoGANs can synthesize mixed emotions (e.g., happiness + surprise).

Identity preservation is implicitly enforced in the encoder, which projects an input image onto a low-dimensional latent

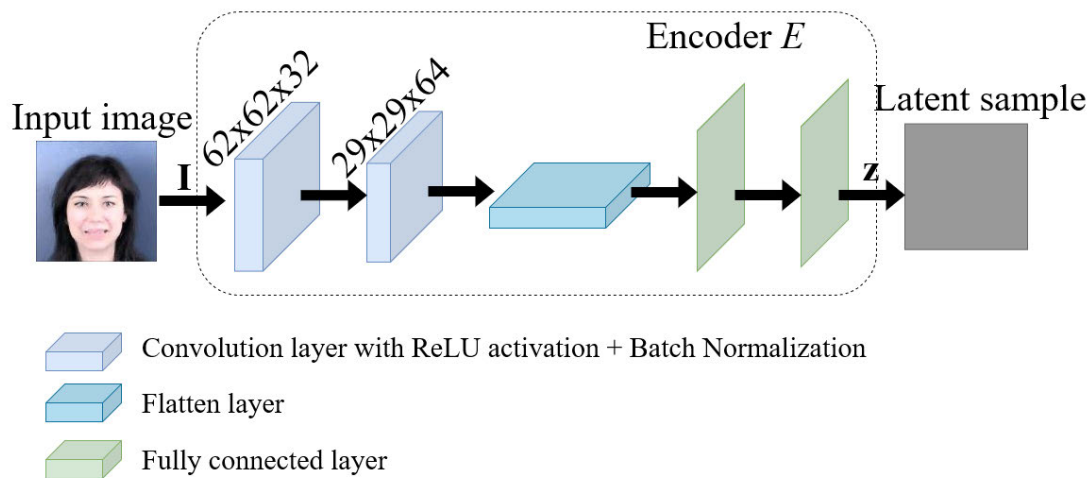


FIGURE 4. Encoder E_z network configuration.

TABLE 1. Evaluation of the encoder's configuration using two metrics: (1) Mean Squared Error (MSE) measures the loss between initial latent(z) and latent predicted by the encoder ($\hat{z}$). (2) Cosine Similarity Score (CSS) measures the similarity of facial features of the generated images from z and $\hat{z}$. VGGFace model is used to extract facial features.

# of Conv block		Two								Three							
Activation Function		All layers: ReLU				Last layer: Linear				All layers: ReLU				Last layer: Linear			
Optimizer		Adam		SGD		Adam		SGD		Adam		SGD		Adam		SGD	
Batch Size		64	128	64	128	64	128	64	128	64	128	64	128	64	128	64	128
JAFPE	MSE	1.03	1.08	1.22	1.32	1.09	1.13	1.27	1.30	1.14	1.16	1.45	1.56	1.28	1.39	1.64	1.67
	CSS	64.6%	63.9%	61.7%	57.6%	91.8%	87.1%	80.8%	81.0%	63.4%	65.4%	41.6%	33.3%	84.9%	78.3%	64.2%	61.1%
MUG	MSE	0.94	1.01	1.14	1.25	0.91	0.99	1.18	1.25	1.10	1.14	1.27	1.35	1.36	1.35	1.33	1.49
	CSS	55.4%	53.5%	56.3%	52.2%	72.8%	65.0%	69.5%	63.3%	52.4%	51.3%	50.4%	45.8%	65.7%	62.4%	60.3%	51.9%

space. These latent samples serve as the core foundation for image generation, ensuring that the primary facial structure from the original image is retained. During image synthesis, while the latent samples maintain the fundamental facial composition, variations in facial expressions are controlled by emotion labels. The identity preservation properties of our proposed method have been validated in section IV-B2.

B. FACIAL EXPRESSION RECOGNITION

The second part introduces the facial expression recognition model for estimating complex emotions using the images generated in the previous part. Complex emotions are defined as a mixture of two or more distinct emotions occurring simultaneously, and they cannot be adequately described by the definitions of basic emotions alone. Therefore, in the previous section, we utilized the multidimensional labels vectors to synthesize facial images that represent complex facial expressions, each generated image containing two basic emotion labels. For example, if an image is synthesized by assigning the respective dimensions for “happiness” and “surprise,” its labels become “happiness” and “surprise.” This method enhances the suitability of a multi-label formulation for emotion estimation.

The network configuration of the emotion estimation model is depicted in Fig. 6. As convolution layers have been proven as effective for image classification [31], [32], [33], we have applied convolution layers to extract facial

expression features. Following the convolution layers, a max-pooling layer is employed to decrease the spatial dimensions of the previously extracted features. These features are subsequently flattened and used for multi-label classification in the final layer, which employs sigmoid activation. To train the FER model, the loss function is based on the standard cross entropy loss to accommodate the sigmoid activation, estimating the presence of facial features associated to each class independently.

Since the model is expected to predict multiple emotion labels, a common metric such as accuracy is an inadequate metric for evaluating multi-label classification. For example, consider an image with the true labels “happiness” and “sadness”, represented as the vector $[0, 0, 0, 1, 1, 0]$, where each dimension corresponds to basic emotions (anger, disgust, fear, happiness, sadness, and surprise). If the model predicts “happiness” and “anger”, resulting in $[1, 0, 0, 1, 0, 0]$, it correctly predicts “happiness” but misclassifies “sadness” as “anger”. Using accuracy in evaluation would treat this prediction as entirely incorrect.

To address this, the f-beta (f_β) metric is employed to evaluate the model. The f_β metric combines precision and recall, weighted by parameter β . Precision assesses how well the model predicts the positive class, while recall measures the model’s ability to identify positive classes. In this multiple-label classification scenario, a specific emotion class from the true label is defined as the positive class, and all

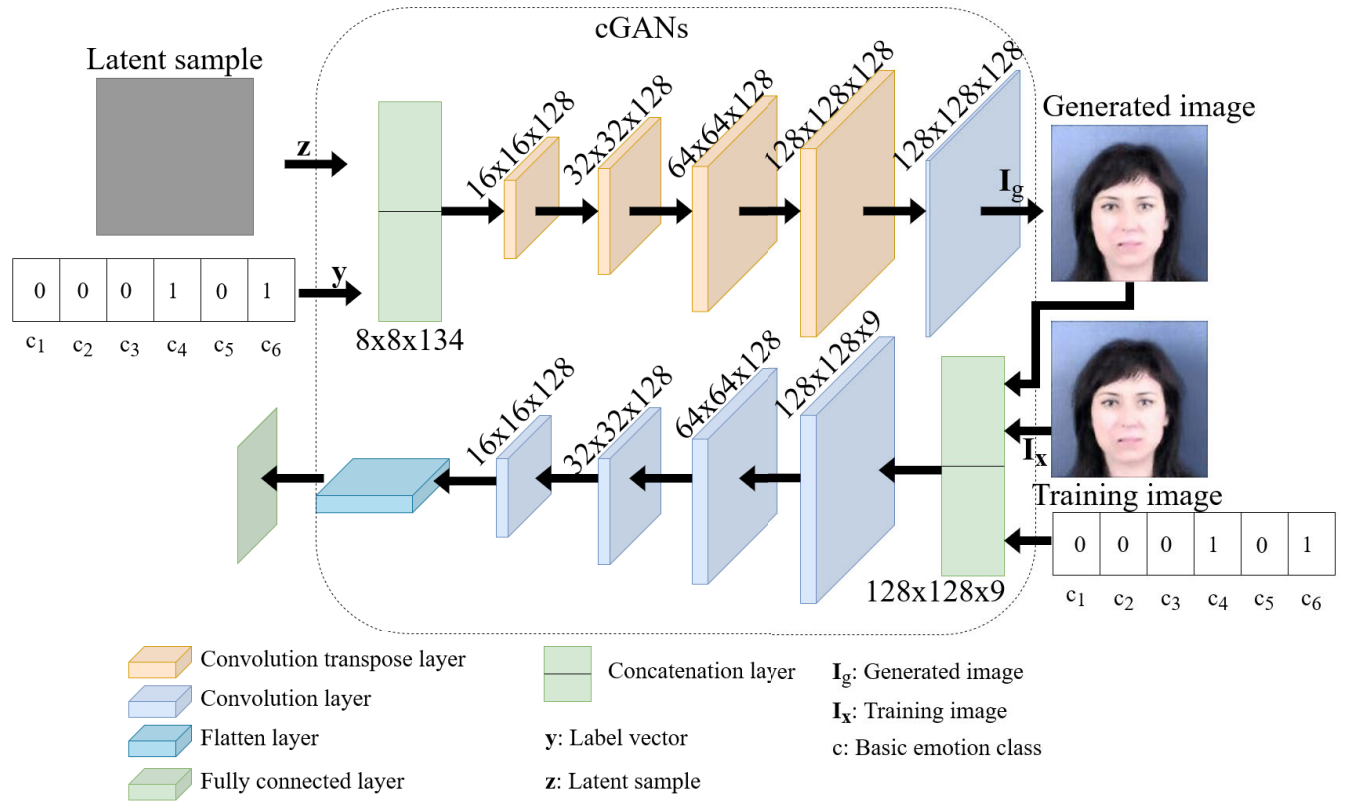


FIGURE 5. Conditional generative adversarial networks (cGANs) configuration.

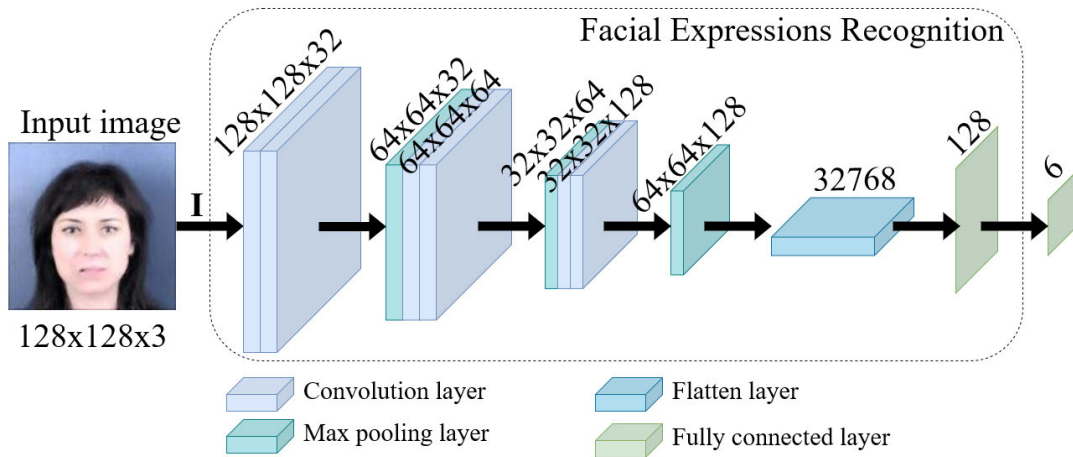


FIGURE 6. Network configuration of a multi-labels estimation model.

others are considered negative classes using a one-versus-the-rest (OVS) approach. The f_β score ranges from 1 (optimal) to 0 (minimal). It is formulated as follows:

$$f_\beta = (1 + \beta^2) \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision} + \text{recall})} \quad (2)$$

where f_β is the f-beta metric which uses weighted parameter β .

IV. EXPERIMENTS

A. DATA PREPARATION AND EXPERIMENTAL SETTING

The experiments make use of the extended Cohn Kanade (CK+) dataset [34], the Japanese female facial expressions (JAFPE) dataset [35] and the Multimedia Understanding Group (MUG) facial expressions dataset [36]. Since this research focuses on analyzing still images of complex facial expressions, the data preparation is required in this research.

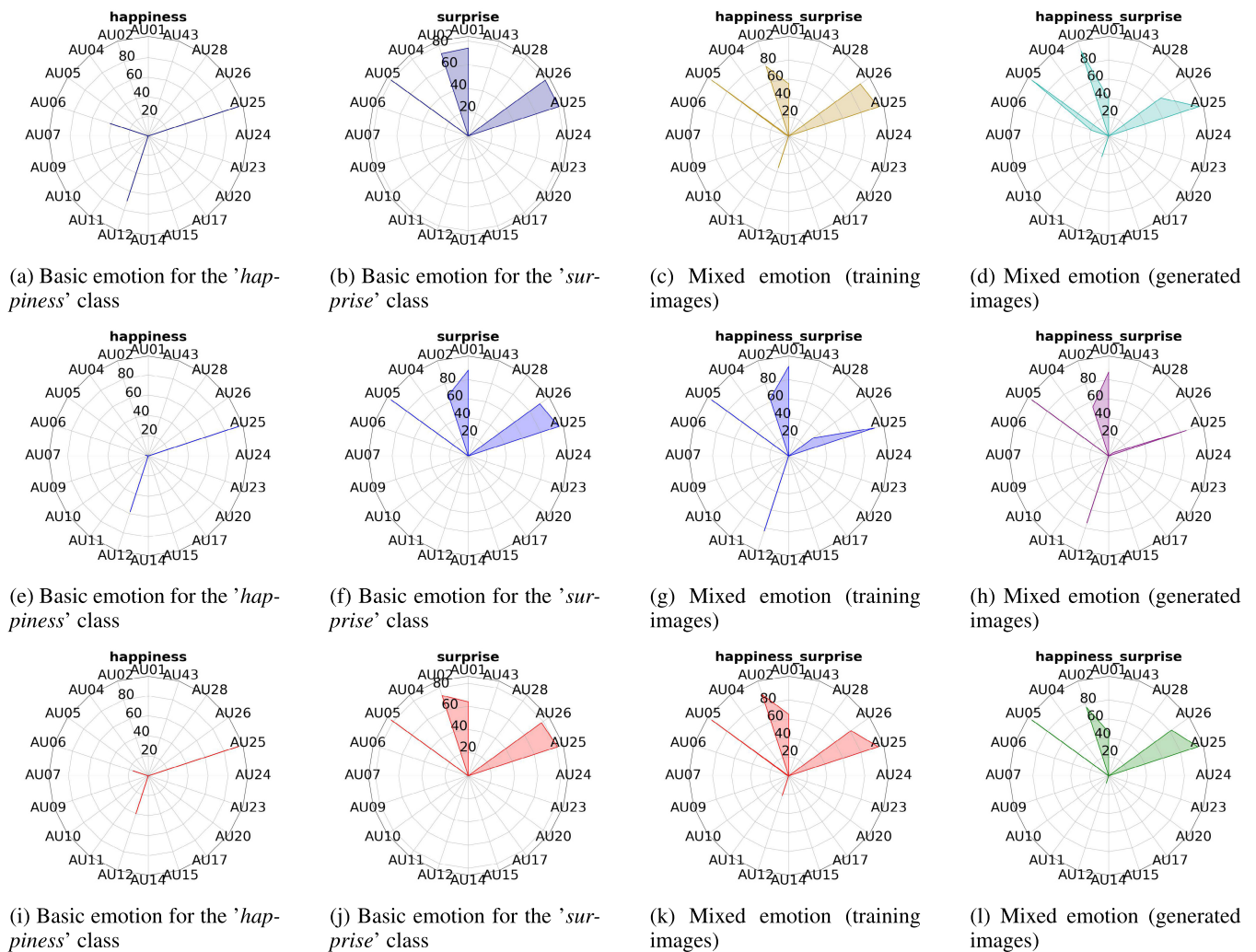


FIGURE 7. Prototypical action units (AUs) for basic emotions, such as “happiness” and “surprise” in this figure, and a combination of emotions among them are depicted. The circular axis represents the AU numbers and the percentage of images with prototypical AUs. The results for the CK+, JAFFE, and MUG datasets are presented accordingly.

For the MUG dataset, which contains sequential images, the methodology described in [37] is employed to select the peak images, which show the most expressive facial expressions for each subject. Following peak image selection, morphing is applied to create complex facial expression images. Morphing combines corresponding local regions in two images with distinct facial expressions to produce an intermediate representation. These resulting images serve as ground truth data and support the training of generative model outlined in Eq. (1). The implementation in this research has been done on the workstation with the CPU Intel i9-9900K, and the GPU NVIDIA RTX A6000. Furthermore, to mitigate the class imbalance issue for the facial expression recognition model, we employed our proposed generative model cEmoGANs for data augmentation to balance the dataset by generating synthetic samples. This can improve the recognition model to learn from a wider distribution of expressions, enhancing its generalization across both frequent and infrequent emotion categories.

B. EVALUATION OF CONDITIONED GENERATIVE ADVERSARIAL NETWORKS (cEmoGANs)

We evaluated the proposed conditioned Generative Adversarial Networks (cEmoGANs) based on two objectives. The primary goal of cEmoGANs is to synthesize complex facial expressions that reflect complex emotions. The second goal is to control facial expressions and facial identity information separately and disentangle the features. Therefore, this section discusses the results obtained from various evaluation perspectives.

1) EVALUATION ON THE ABILITY TO GENERATE MIXED FACIAL EXPRESSION IMAGES

To confirm the presence of complex facial expressions in the generated images, we utilize the method outlined in [7], which involves Action Unit (AU) recognition. For instance, if a facial expression that reflects a combination of happiness and surprise is found in the generated images, the corresponding AUs for each emotion (happiness and surprise)

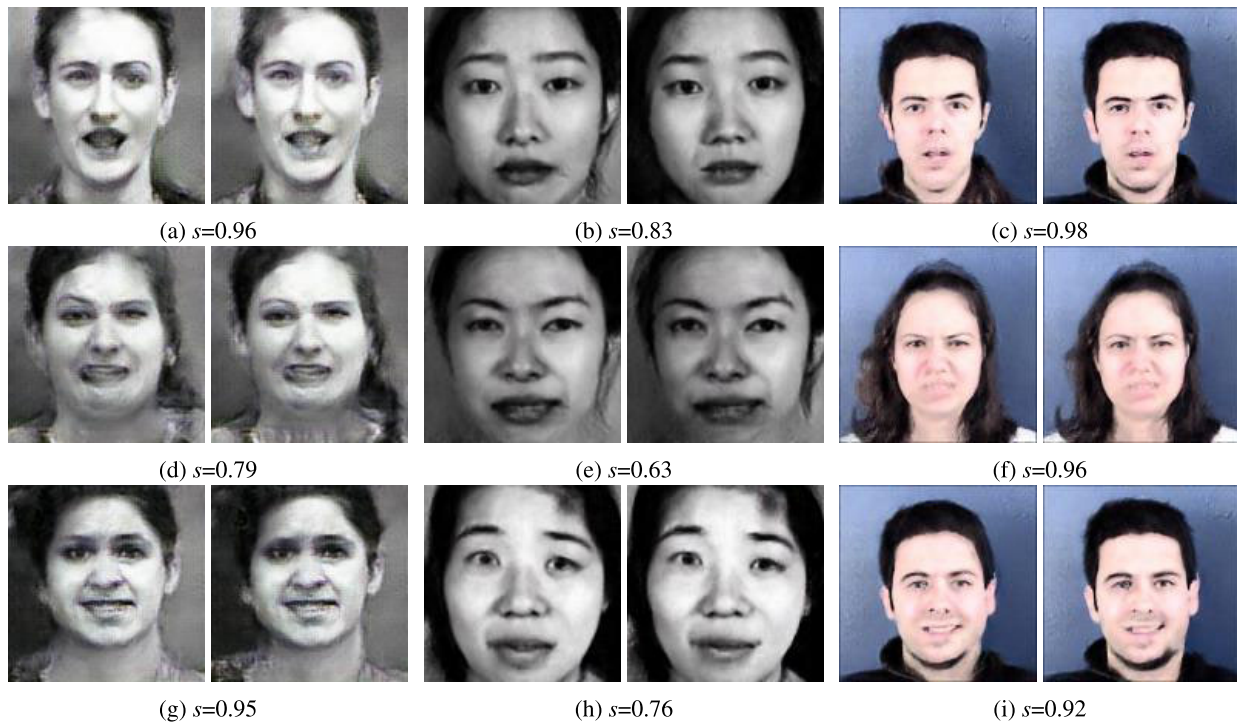


FIGURE 8. Experimental results for evaluation of identity preservation by Encoder. s is the cosine similarity score between the facial representation of the image from the initial latent (left) and reconstructed image from estimated latent $\hat{z}$ (right) for each pair. The score ranges from +1 (most similarity) to -1 (most dissimilarity).

should be activated in the observed image, aligning with the AUs in the training images.

We identify the prototypical Action Units (AUs) for each basic emotion using a pre-trained recognition model [38] and select the relevant AUs for mixed emotions. In Fig. 7, the standard AUs for happiness include AU06, AU12, and AU25, while the AUs for surprise consist of AU01, AU02, AU05, AU25, and AU26. The AUs of interest for the mixed emotion “happily surprised” represent the union of two AU sets to each individual emotion. Fig. 7 illustrates the number of images in which these interest AUs are activated. For example, 60% of the images in JAFFE activated AU12, while fewer than 20% display AU05 to express the emotion of happiness in Fig. 7(e). A similar interpretation applies to all the charts.

In Fig. 7, the first two columns display the distribution of images with activated Action Units (AUs) for basic emotions, the third column presents morphed images with complex facial expressions used as training data in adversarial training, and the last column exhibits generated images reflecting complex emotions. The first row represents the experimental results for the CK+ dataset.

From Fig. 7(a), it is evident that individuals frequently use AU12 and AU25 to express happiness, whereas AU6 is less commonly used. In contrast, for surprise expressions as shown in Fig. 7(b), all relevant AUs are activated in at least 70% of CK+ images. The training images resulting from the morphing process, which combines images with

happiness and surprise expressions, exhibit a minor activation of AU6 and AU12 but a major activation of AU5 and AU25. Since morphed images directly combine facial components of the two expressions, they are guaranteed to exhibit mixed facial expressions. Comparing the generated images with the morphed ones confirms that the generated images also capture mixed facial expressions. Fig. 7(d) exhibits a similar pattern, with AU6 and AU12 being less commonly activated AUs, while AU5 and AU25 are more frequently activated.

The second row in Fig. 7 corresponds to the results for the JAFFE dataset. Similar to CK+, AU06 is the least activated AUs, with less than 10% of individuals using AU06 to express happiness, as shown in Fig. 7(e). For surprise expressions in Fig. 7(f), AU02 is the least frequently activated AU compared to the other AUs, with 65% of individuals employing it. When combining these two expressions through morphing, AU25 becomes one of the least activated AUs in Fig. 7(g). The same effect can be found in the chart featuring the generated images in Fig. 7(h).

The last row in Fig. 7 presents the results for the MUG dataset. Once again, we observe that AU06 is among the least activated AUs, as depicted in Fig. 7(i). These results lead to the conclusion that the majority of individuals primarily use AU12 and AU25 to express happiness. The chart for surprise expressions resembles the CK+ findings. In MUG, AU02 is more commonly used compared to AU01, which differs from JAFFE, where AU02 is less frequently employed



FIGURE 9. Example output of the label swapping to evaluate features disentanglement. G is the generator of cEmoGANs. z_1 and y_1 are the first set of randomly generated latent sample and label vector, whereas z_2 and y_2 are the second set. $G(z, y)$ indicates that the latent sample is mapped onto image space by the generator G . The rows are arranged in the order of CK+, JAFFE and MUG respectively.

than AU01. These findings suggest that Japanese individuals are less inclined to raise their outer brow (AU02) to convey surprise. The results for morphed images in Fig. 7(k) reveal that AU06 and AU12 are among the least activated AUs, and similar results are seen in the generated images shown in Fig. 7(l). In summary, the distributions of activated AUs in the generated images closely resemble those of the morphed images. Consequently, we can conclude that the proposed cEmoGANs model successfully synthesizes complex facial expressions to convey complex emotions.

2) EVALUATION ON IDENTITY PRESERVATION

To ascertain whether encoder E preserves identity in the generated images, the process begins by creating random latent samples z with corresponding labels y . These vectors are then utilized by G to generate the image I_g , which is subsequently passed to E for facial encoding. After that, E estimates the latent sample $\hat{z}$, which is subsequently mapped back to the image space by the same generator, G . A pair of generated images, one based on the initial latent sample z and the other on the estimated latent sample $\hat{z}$, are provided to a pre-trained face recognition model by [31] for extracting facial embeddings. The cosine similarity among these facial features is then measured.

The results are presented in Fig. 8. It is evident that the generated images, which utilize the encoded latent sample from the encoder (E), retain the original face identity and

maintain facial resemblance while exhibiting subtle changes, such as variations in hairlines are smoothed out in Fig. 8(f) and (i). We use a threshold value of 0.7 to assess the similarity of the test set of entire generated dataset and achieve relatively high performance in maintaining identity through the encoder. Specifically, 98% of the generated images, employing encoded samples from the test set in CK+, exhibit high similarity scores, along with 94% in JAFFE and 99% in MUG.

3) EVALUATION ON FEATURE DISENTANGLEMENT

The proposed cEmoGANs independently define the features for facial expressions and identity information, thus disentangling these features in the latent space. To assess the impact of this property, we adopt the method presented in [39] to determine how the label vector encodes facial expressions in the generated image. By considering two sets of latent and label pairs (z_1, y_1 and z_2, y_2), we observe that swapping the labels between the latents can interchange the corresponding facial expressions.

The results of label swapping can be observed in Fig. 9. The first two columns depict the outcomes of two sets of generated images using random latent samples and label vectors, while the last two columns display the results of label swapping. Despite being micro-expressions, noticeable changes can be observed as the labels are swapped, as depicted in Fig. 9. The results for CK+ and MUG exhibit distinct alterations

TABLE 2. The inception score results, which assess the quality and diversity of the generated images, for the proposed cEmoGANs.

Models	Datasets		
	CK+	JAFPE	MUG
GANs [40]	5.7 (± 0.4)	9.9 (± 0.4)	3.8 (± 0.1)
DCGANs [42]	8.7 (± 0.5)	10.1 (± 0.4)	8.4 (± 0.2)
StyleGANs [43]	8.7 (± 0.6)	5.0 (± 0.3)	11.4 (± 0.2)
WGANs [44]	5.9 (± 0.3)	3.9 (± 0.3)	7.6(± 0.2)
FastGANs [45]	6.0 (± 0.6)	9.2 (± 0.5)	6.0 (± 0.1)
EmoGANs [46]	8.9 (± 0.6)	4.4 (± 0.3)	8.2 (± 0.2)
EmoGANs1 [47]	10.1 (± 0.5)	8.4 (± 0.4)	4.1 (± 0.1)
EmoGANs2 [48]	9.7 (± 0.9)	8.8 (± 0.4)	8.8 (± 0.3)
Proposed model	8.8 (± 0.6)	9.0 (± 0.2)	11.4 (± 0.1)

TABLE 3. The Frechet inception distance results, which evaluate the quality of generated images, for the proposed cEmoGANs.

Models	Datasets		
	CK+	JAFPE	MUG
GANs [40]	254.2	242.0	559.4
DCGANs [42]	53.8	143.8	182.7
StyleGANs [43]	98.3	647.7	71.0
WGANs [44]	219.0	1102.1	98.1
FastGANs [45]	205.8	81.4	934.9
EmoGANs [46]	57.5	713.9	385.9
EmoGANs1 [47]	72.6	281.4	1140.3
EmoGANs2 [48]	25.0	309.8	467.4
Proposed model	56.4	107.8	136.6

TABLE 4. The blind/referenceless image spatial quality evaluator metric results, which quantify distortion in generated images, for the proposed cEmoGANs.

Models	Datasets		
	CK+	JAFPE	MUG
GANs [40]	30.7	20.8	35.4
DCGANs [42]	28.4	27.9	63.3
StyleGANs [43]	19.7	21.8	16.5
WGANs [44]	34.8	61.2	28.8
FastGANs [45]	72.1	75.3	38.4
Morphing	27.6	25.7	18.9
EmoGANs [46]	29.4	85.0	53.8
EmoGANs1 [47]	25.4	54.8	155.3
EmoGANs2 [48]	29.1	30.8	25.0
Proposed model	20.1	31.8	11.0

in their facial expressions. These findings provide evidence that the inclusion of an encoder and multi-dimensional labels effectively disentangles facial features and expressions in the image generation process.

4) EVALUATION ON THE QUALITY OF GENERATED IMAGES

The quality of generated images is crucial for subjective assessments. To assess image quality, we will mainly discuss on a set of metrics commonly employed for evaluating images generated by GANs.

a: INCEPTION SCORE

The Inception score, initially introduced by Salimans et al. [49], provides an automated method for assessing the quality of generated images, aiming to reduce the reliance on subjective evaluations. This metric incorporates the Inception model [50], which was originally trained for image classification tasks, to classify the generated images. The

Inception score evaluates two key aspects of the generated images: image quality and diversity.

In terms of image quality, the score assesses whether the generated images closely resemble objects that the pre-trained InceptionV3 model can classify. High-quality images exhibit a low entropy value, indicating that their confidence probability scores are concentrated on a single value. Diversity, on the other hand, is evaluated based on the variety of images the generator can produce. A higher diversity leads to higher entropy values. The Inception score ranges from 1 to 1000, with a higher score indicating better performance. It is a valuable metric that effectively combines image quality and diversity assessment.

The Inception Score metric calculates the conditional probability $p(y|x)$ to gauge image quality, which measures the confidence probability of an image being a particular class label according to the recognition model. To assess diversity among the images, it employs the KL divergence, measuring the relative entropy between the conditional probability and

TABLE 5. Evaluation of the proposed complex facial expression recognition model on the test set using two metrics: (1) f_β , the f-beta metric defined in Eq.(2), which balances precision and recall; (2) EMR, which calculates the proportion of the predicted label sets that exactly match the true label sets.

Models	Metrics					
	f_β			EMR		
	CK+	JAFFE	MUG	CK+	JAFFE	MUG
EfficientNet B0 [54]	0.45	0.60	0.51	6.2%	5.0%	5.9%
Proposed emotion estimation model (StyleGANs [43])	0.42	0.54	0.53	14.2%	25.6%	26.0%
Proposed emotion estimation model (cEmoGANs)	0.99	0.99	0.80	100%	99%	68.7%

the marginal probability, which represents the probability distribution of all generated images. A high Inception Score signifies both high image quality and diversity, making it a useful tool for evaluating generated images.

However, it is important to note that the Inception score is applicable only to images containing objects recognized by the model used in its calculation, such as the Inception model. The Inception model is primarily trained for object classification and not designed for classifying facial expressions. Therefore, we opted to use a facial expression recognition model instead. We employed a standard four-layered convolutional neural network using the morphed images to recognize complex facial expressions. The evaluation of the model was conducted on 10-fold cross-validated datasets. Based on the experimental results, we achieved recognition performance of approximately 73% in CK and 98% in JAFFE and MUG.

We employed a facial expressions model to compute the Inception Score, which classifies 15 types of complex emotions and yields a score ranging from 1 to 15. The experimental results are presented in Table. 2 and are compared to other models. The quality of images generated by the proposed cEmoGANs ranked first in MUG and third in CK+ and JAFFE. Compared to other EmoGANs models, the proposed cEmoGANs ranked first in two out of three datasets such as JAFFE and MUG. This result shows that adding control in disentangling the features improves the quality of generated images and enhances data diversity.

b: FRECHET INCEPTION DISTANCE

Heusel et al. [51] introduced the Frechet Inception Distance (FID) as a metric to assess the performance of generative models. In contrast to the previous Inception Score metric, which do not consider training images, the FID metric was designed to measure the similarity between generated and training images. To achieve this, FID employs a pre-trained inception model as a feature extractor to capture feature representations of both the generated and training images. These features are then summarized as multivariate Gaussian distributions, and the similarity between these distributions is quantified using either the Frechet distance or the Wasserstein-2 distance. The FID metric provides a measure of the similarity between the generated and training image distributions, offering a more comprehensive evaluation of generator performance.

Similar to the inception score metric, the calculation for FID uses previous facial expressions model instead of the inception model. As FID is a distance-based metric, its scores range from zero, indicating the highest similarity between the training and generated distributions. The results are presented in the Table.3. The proposed model is ranked second in JAFFE and third in CK+ and MUG. Compared to other EmoGANs models, the proposed cEmoGANs is ranked first in JAFFE and MUG, consistent with the results from the Inception Score metric. Results from the FID metric demonstrate that the distributions between generated images and training images have a closer distance and are similar.

c: BLIND/REFERENCELESS IMAGE SPATIAL QUALITY EVALUATOR (BRISQE)

Mittal et al. [52] introduced the BRISQE metric for evaluating distortions in images without a reference image. The metric quantifies distortion by measuring the deviation of the normalized distorted images from the standard Gaussian distribution, which is typically followed by high-quality images. In the BRISQE computation, the input images are normalized and oriented at different angles to compute intensity features by fitting them to the Generalized Gaussian Distribution (GGD). These features are subsequently employed by a Support Vector Machine (SVM) to generate a quality score. A lower score indicates less distortion in the image and corresponds to higher image quality.

The BRISQE results are presented in Table.4. Unlike the previous two metrics, Inception Score and Frechet Inception Distance, evaluation of morphing images is included in BRISQE metric. The rationale for this inclusion is that ground truth images are essential for computing the previous metrics for comparison, and morphing images have been utilized both as ground truth images and for training the recognition model. As BRISQE does not require the reference images for comparison, it is possible to evaluate the quality of morphing images.

Based on the calculated results, the generated images by the proposed model using MUG datasets exhibit the least distortions among the images generated by different generators. The proposed model ranked second with CK+, which is an improvement over the morphed images used as ground truth during adversarial training. With JAFFE data, the proposed model ranked fifth, but its scores are the best among the previous EmoGANs models. Among other EmoGANs variations, the current proposed model provides the best score in two out of three datasets (CK+ and MUG).

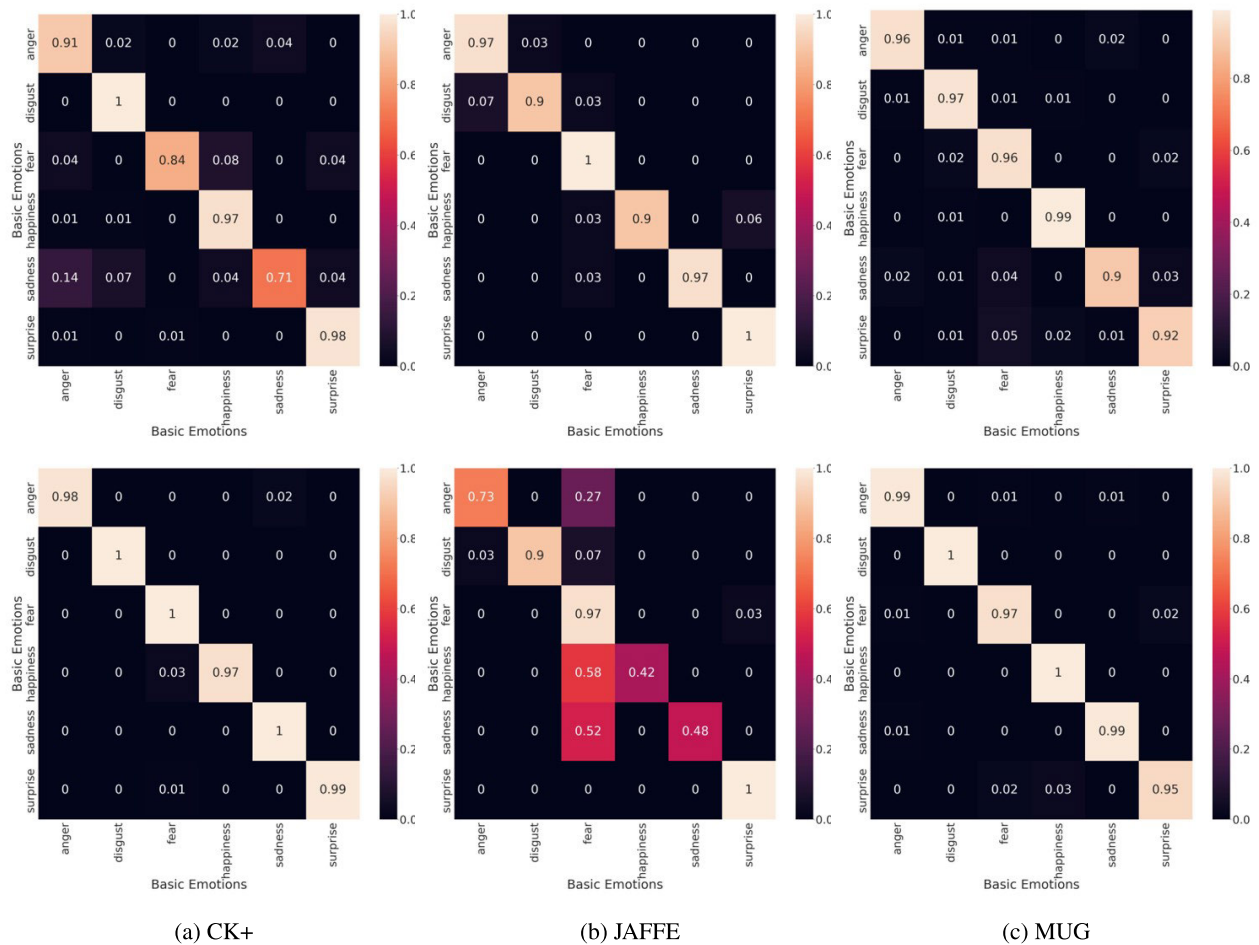


FIGURE 10. Evaluation for basic emotions recognition by the proposed model (upper row) and the baseline method (EfficientNet B0) (lower row).

This suggests that the current configuration improves the quality of generated images by producing less distorted images.

C. EVALUATION OF COMPLEX FACIAL EXPRESSION RECOGNITION MODEL

In this section, we assess the complex emotion estimation model by recognizing the emotions conveyed in complex facial expressions within the generated images produced by the proposed conditioned EmoGANs model. As discussed earlier in the preceding section, the f_β metric is more suitable than the accuracy metric for multi-labels classification. In addition, we have included the exact match ratio (EMR) to measure the proportion of cases where the prediction output precisely matches the ground truth labels.

The performance of the proposed model is compared with two other models. It has been demonstrated that StyleGANs can mix different styles from two images [43]. Therefore, we assess the network configuration of the proposed model using images generated by the StyleGANs generator. As reported in the study by Khine et al. [53], the EfficientNet B0 model was found to be the best-performing model for recognizing basic facial expressions. Consequently,

we incorporated the EfficientNet B0 and trained it using images generated by cEmoGANs. For the data, 15,000 generated images by cEmoGANs and StyleGANs are divided into training and test sets at a 7:3 ratio. The results are presented in Table 5.

In Table 5, the f_β metric calculates partially correct labels, whereas the EMR metric accounts for exact match predictions. Consequently, the scores returned by EMR are significantly lower than the scores obtained by f_β . Based on the results obtained from the two metrics, the proposed network configuration performs better when using images generated by the proposed cEmoGANs than when using images generated by StyleGANs in recognizing the facial expressions for estimating complex emotions. This is because cEmoGANs is specifically designed to synthesize images of complex facial expressions using label vectors.

While the model outperforms the compared models in all datasets, its effectiveness is somewhat diminished when applied to the MUG dataset. There are two possible reasons for this. The first reason is distinct characteristics of the datasets. The second reason is that the original data in the MUG dataset is presented in the form of image sequences. During data preparation, peak frames are estimated and

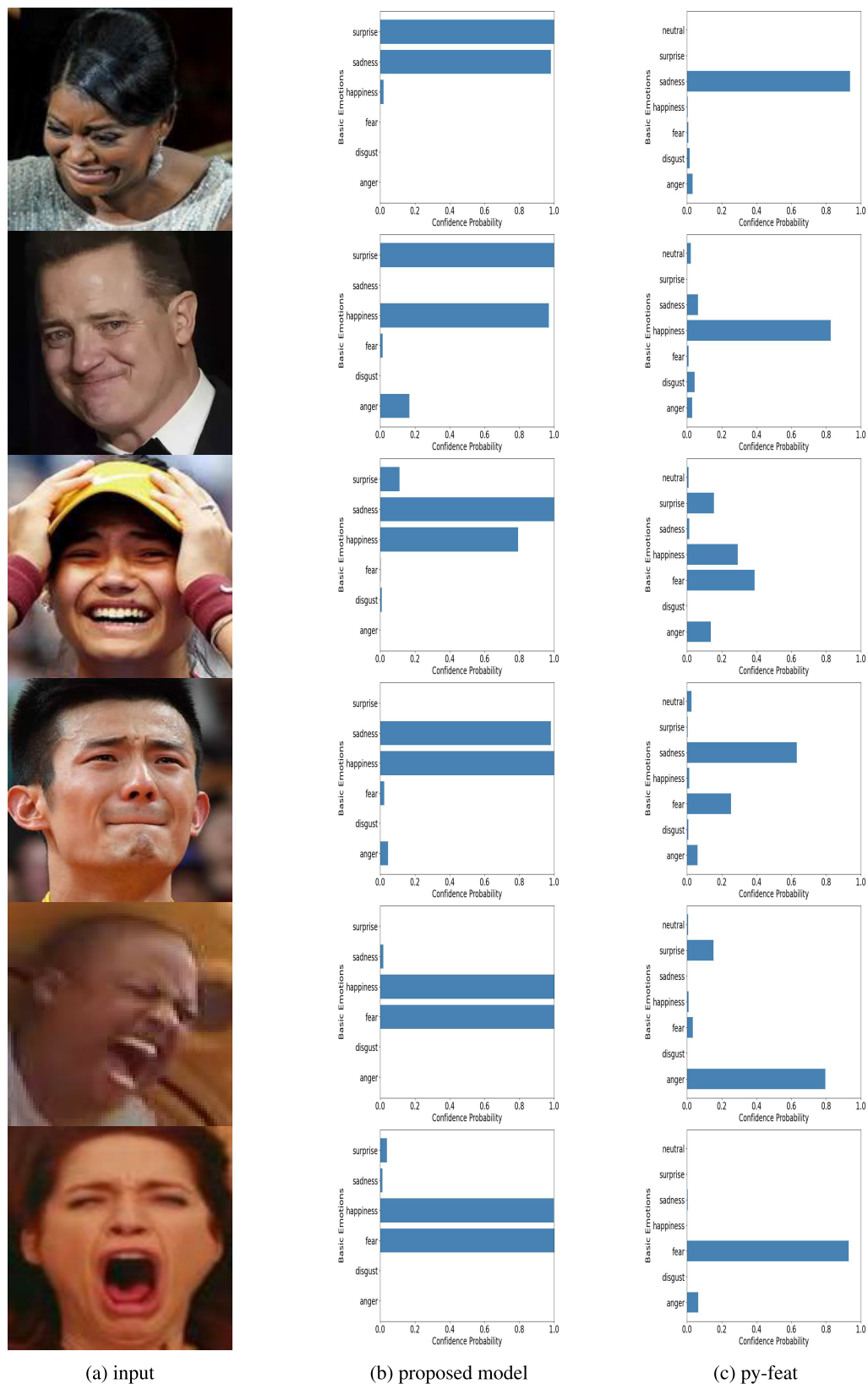


FIGURE 11. Estimation of mixed emotions of unconstrained images [55], [56], [57], [58], [59], [60] by the proposed model and Py-feat [38].

selected. Consequently, some of the chosen peak frames may not capture fully expressive facial expressions, as seen in the CK+ and JAFFE datasets. Therefore, labels of 69% of the test images are correctly predicted by the model.

We also compared the proposed model with EfficientNet B0, which is recognized as the best model for recognizing basic facial expressions in [53]. According to the results in Table. 5, although EfficientNet B0 demonstrates reliable performance in the estimation of basic emotions, it is not suitable for estimating complex emotions. Therefore, we evaluated the proposed model for basic emotions estimation. To assess performance, the proposed model and EfficientNet B0 were trained and tested using basic facial expression images.

Results are depicted in Fig. 10. In the CK+ dataset results, the proposed model misinterprets sadness expressions as anger and fear as happiness, whereas EfficientNet B0 achieves almost perfect scores. In the JAFFE dataset, the proposed model shows better recognition than EfficientNet B0. In MUG, the scores of the two models are comparable.

We also conducted tests with the proposed model using unseen images from Fig. 1 and compared the results with Py-Feat facial expression recognition model [38]. Results are shown in Fig. 11. The first two images are taken at the awarding moment of the Oscar award. Although it is the happiness event, their faces can be perceived as the sadness because of overly happiness. The second and third images are taken at the winning moment of the Olympic games. Although they won the games, their emotions are not just happiness. The last two images capture the facial expressions of the persons who are riding the roller-coaster. Riding roller-coaster is for fun and happiness but they feel scared during riding. Based on the estimated results, the proposed model provides not only the primary emotion but also the secondary emotion which matches the context of the images.

V. CONCLUSION

In this research, we take a step forward in addressing the limitations of traditional emotion classification by redefining emotion classes as compound expressions that extend beyond basic emotions. While this concept is straightforward, it has been underexplored even in recent studies, as many approaches rely on the simplified 6- or 7-class classification framework, which makes implementation easier but oversimplifies emotional expression.

To overcome this limitation, we employ an *analysis-by-synthesis* approach that leverages the synergy between generation and recognition. By synthesizing compound facial expressions, our method provides a broader and more expressive representation of facial expression features. Furthermore, this approach helps mitigate the challenge of limited labeled data by generating complex expressions for training the recognition model, improving its ability to handle diverse emotional expressions.

The proposed approach consists of two main components: facial image generation and emotion estimation. In the

first part, we introduce Conditioned Generative Adversarial Networks (cEmoGANs). Unlike existing models in the literature, the proposed cEmoGANs can preserve facial identity information in generated images, and provide control over the types of complex emotions. It also improves image quality, enhances data diversity, and results in less distorted images as shown in the experimental results.

In the second part, a simple convolutional neural network is employed to recognize complex facial expressions from the images generated by cEmoGANs for complex emotion estimation. Given that complex emotions are defined as the combination of two or more basic emotions, our model is formulated using multi-label classification and evaluated using f_β and exact match ratio.

Evaluation results show that the facial expression recognition model trained on images generated by the proposed cEmoGANs demonstrated significantly better performance than the model with the same network configuration but trained on images generated by other generative methods. Additionally, the recognition model was compared with the state-of-the-art model in basic facial expression recognition tasks, and achieves competitive performance in those tasks.

To further assess generalization on complex facial expression, the model was evaluated using unconstrained facial images that were completely unseen during training. The model's output aligned with the emotions inferred from the context. Although our experiments demonstrate that the model handles unseen images well, we anticipate that biases may exist due to dataset limitations. One of the most vital factors is the demographic diversity of facial expressions across ethnicities in multicultural societies [61], [62]. Furthermore, diversity across different individuals, such as age and gender, might be an influential factor in model performance. As affective computing systems are increasingly applied across diverse populations, addressing these diversity challenges is essential.

Moreover, the backbone deep neural network architecture can be further improved to enhance the performance of the proposed method. For instance, integrating more efficient activation functions, such as the Weight Initialization-Based Rectified Linear Unit (WiB-ReLU) [63], refine optimization techniques, and incorporate advanced training strategies. It is also important to further explore how different backbone architectures make decisions, as the interpretability of the features learned by convolutional neural networks for facial expression recognition of complex emotion may vary accordingly.

Lastly, while facial expressions play a critical role in conveying emotions, they alone may be insufficient to fully capture the full spectrum of human emotions. Since real-world emotional expressions are multimodal, incorporating additional cues such as tone of voice, body language, and physiological signals could significantly enhance emotion interpretation. These aspects are expected to be thoroughly investigated in future research.

APPENDIX

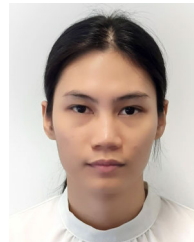
ETHICAL AND PRIVACY STATEMENT

Both the generative and recognition models in this research are trained on publicly available datasets. However, if the proposed method is applied on a larger scale for real-world applications, ethical guidelines, such as informed consent and data protection, must be strictly followed. In our method, the generative model is solely used within the training environment of the recognition model. The generated images are not employed outside the training phase, ensuring that only the recognition model is deployed in real-world applications. As a result, the proposed method does not introduce privacy concerns, as no synthetic facial images are used beyond the training scope.

REFERENCES

- [1] A. Mehrabian and S. R. Ferris, "Inference of attitudes from nonverbal communication in two channels," *J. Consulting Psychol.*, vol. 31, no. 3, pp. 248–252, 1967.
- [2] D. Matsumoto and B. Willingham, "Spontaneous facial expressions of emotion of congenitally and noncongenitally blind individuals," *J. Personality Social Psychol.*, vol. 96, no. 1, pp. 1–10, Jan. 2009.
- [3] J. A. Russell, "A circumplex model of affect," *J. Personality Social Psychol.*, vol. 39, no. 6, pp. 1161–1178, Dec. 1980.
- [4] H. Zhang, J. Yin, and X. Zhang, "The study of a five-dimensional emotional model for facial emotion recognition," *Mobile Inf. Syst.*, vol. 2020, pp. 1–10, Dec. 2020.
- [5] P. Ekman and W. V. Friesen, "Constants across cultures in the face and emotion," *J. Personality Social Psychol.*, vol. 17, no. 2, pp. 124–129, 1971.
- [6] S. Du, Y. Tao, and A. M. Martinez, "Compound facial expressions of emotion," *Proc. Nat. Acad. Sci. USA*, vol. 111, no. 15, pp. E1454–E1462, Apr. 2014.
- [7] S. Du and A. M. Martinez, "Compound facial expressions of emotion: From basic research to clinical applications," *Dialogues Clin. Neurosci.*, vol. 17, no. 4, pp. 443–455, Dec. 2015.
- [8] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.
- [9] M. Ranzato, J. Susskind, V. Mnih, and G. Hinton, "On deep generative models with applications to recognition," in *Proc. CVPR*, Colorado Springs, CO, USA, Jun. 2011, pp. 2857–2864.
- [10] S. Rifai, Y. Bengio, A. Courville, P. Vincent, and M. Mirza, "Disentangling factors of variation for facial expression recognition," in *Proc. 12th Eur. Conf. Comput. Vis. (ECCV)*, Florence, Italy, Jan. 2012, pp. 808–822.
- [11] P. Liu, S. Han, Z. Meng, and Y. Tong, "Facial expression recognition via a boosted deep belief network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 1805–1812.
- [12] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [13] B. Olimov, B. Subramanian, R. A. A. Ugli, J.-S. Kim, and J. Kim, "Consecutive multiscale feature learning-based image classification model," *Sci. Rep.*, vol. 13, no. 1, pp. 1–16, Mar. 2023.
- [14] B. A. U. Olimov, K. C. Veluvolu, A. Paul, and J. Kim, "UzADL: Anomaly detection and localization using graph Laplacian matrix-based unsupervised learning method," *Comput. Ind. Eng.*, vol. 171, Sep. 2022, Art. no. 108313.
- [15] V.-T. Dang, H.-Q. Do, V.-V. Vu, and B. Yoon, "Facial expression recognition: A survey and its applications," in *Proc. 23rd Int. Conf. Adv. Commun. Technol. (ICACT)*, Pyeong Chang, South Korea, Feb. 2021, pp. 359–367.
- [16] D. Hamster, P. Barros, and S. Wermter, "Face expression recognition with a 2-channel convolutional neural network," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Killarney, Ireland, Jul. 2015, pp. 1–8.
- [17] L. Nwosu, H. Wang, J. Lu, I. Unwala, X. Yang, and T. Zhang, "Deep convolutional neural network for facial expression recognition using facial parts," in *Proc. IEEE 15th Intl Conf Dependable, Autonomic Secure Comput., 15th Intl Conf Pervasive Intell. Comput., 3rd Intl Conf Big Data Intell. Comput. Cyber Sci. Technol. Congress (DASC/PiCom/DataCom/CyberSciTech)*, Orlando, FL, USA, Nov. 2017, pp. 1318–1321.
- [18] B. Chakravarthi, S.-C. Ng, M. R. Ezilarasan, and M.-F. Leung, "EEG-based emotion recognition using hybrid CNN and LSTM classification," *Frontiers Comput. Neurosci.*, vol. 16, Oct. 2022, Art. no. 1019776.
- [19] L. Wang, J. Hao, and T. H. Zhou, "ECG multi-emotion recognition based on heart rate variability signal features mining," *Sensors*, vol. 23, no. 20, p. 8636, Oct. 2023.
- [20] Z. Huang, M. Dong, Q. Mao, and Y. Zhan, "Speech emotion recognition using CNN," in *Proc. 22nd ACM Int. Conf. Multimedia*, New York, NY, USA, Nov. 2014, pp. 801–804.
- [21] B. Subramanian, B. Olimov, S. M. Naik, S. Kim, K.-H. Park, and J. Kim, "An integrated mediapipe-optimized GRU model for Indian sign language recognition," *Sci. Rep.*, vol. 12, no. 1, Jul. 2022, Art. no. 11964.
- [22] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [23] Z. Yu, G. Liu, Q. Liu, and J. Deng, "Spatio-temporal convolutional features with nested LSTM for facial expression recognition," *Neurocomputing*, vol. 317, pp. 50–57, Nov. 2018.
- [24] F. An and Z. Liu, "Facial expression recognition algorithm based on parameter adaptive initialization of CNN and LSTM," *Vis. Comput.*, vol. 36, no. 3, pp. 483–498, Mar. 2020.
- [25] M. Usman, S. Latif, and J. Qadir, "Using deep autoencoders for facial expression recognition," in *Proc. 13th Int. Conf. Emerg. Technol. (ICET)*, Islamabad, Pakistan, Dec. 2017, pp. 1–6.
- [26] N. Zeng, H. Zhang, B. Song, W. Liu, Y. Li, and A. M. Dobaie, "Facial expression recognition via learning deep sparse autoencoders," *Neurocomputing*, vol. 273, pp. 643–649, Jan. 2018.
- [27] B. Olimov, J. Kim, and A. Paul, "REF-net: Robust, efficient, and fast network for semantic segmentation applications using devices with limited computational resources," *IEEE Access*, vol. 9, pp. 15084–15098, 2021.
- [28] B. Olimov, S.-J. Koh, and J. Kim, "AEDCN-net: Accurate and efficient deep convolutional neural network model for medical image segmentation," *IEEE Access*, vol. 9, pp. 154194–154203, 2021.
- [29] A. Larsen, S. K. Sønderby, H. Larochelle, and O. Winther, "Autoencoding beyond pixels using a learned similarity metric," in *Proc. Int. Conf. Mach. Learn. (ICML)*, New York, NY, USA, Jun. 2016, pp. 1558–1566.
- [30] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.
- [31] O. Parkhi, A. Vedaldi, and A. Zisserman, "Deep face recognition," in *Proc. Brit. Mach. Vis. Conf. (BMVC)*, Jan. 2015, pp. 41.1–41.12.
- [32] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
- [33] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 4510–4520.
- [34] P. Lucey, J. F. Cohn, T. Kanade, J. Saragih, Z. Ambadar, and I. Matthews, "The extended cohn-kanade dataset (CK+): A complete dataset for action unit and emotion-specified expression," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, San Francisco, CA, USA, Jun. 2010, pp. 94–101.
- [35] M. Lyons, S. Akamatsu, M. Kamachi, and J. Gyoba, "Coding facial expressions with Gabor wavelets," in *Proc. 3rd IEEE Int. Conf. Autom. Face Gesture Recognit.*, Nara, Japan, Jun. 1998, pp. 200–205.
- [36] N. Aifanti, C. Papachristou, and A. Delopoulos, "The MUG facial expression database," in *Proc. 11th Int. Workshop Image Anal. Multimedia Interact. Services WIAMIS 10*, Desenzano del Garda, Italy, Apr. 2010, pp. 1–4.
- [37] W. S. S. Khine, P. Siritanawan, and K. Kotani, "Automatic peak frame selection from dynamic facial expressions," in *Proc. 60th Annu. Conf. Soc. Instrum. Control Eng. Jpn. (SICE)*, Tokyo, Japan, Sep. 2021, pp. 1088–1093.
- [38] J. H. Cheong, E. Jolly, T. Xie, S. Byrne, M. Kenney, and L. J. Chang, "Pyfeat: Python facial expression analysis toolbox," *Affect. Sci.*, vol. 4, no. 4, pp. 781–796, Dec. 2023.

- [39] G. Perarnau, J. V. D. Weijer, B. Raducanu, and J. M. Álvarez, "Invertible conditional GANs for image editing," in *Proc. NIPS Workshop Adversarial Training*, Jan. 2016, pp. 1–18.
- [40] R. Labaca-Castro, "Generative adversarial nets," in *Proc. Neural Inf. Process. Syst. (NIPS)*, Jan. 2023, pp. 73–76.
- [41] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, Oct. 2020.
- [42] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," 2015, *arXiv:1511.06434*.
- [43] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 12, pp. 4217–4228, Dec. 2021.
- [44] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, Jul. 2017, pp. 214–223.
- [45] B. Liu, Y. Zhu, K. Song, and A. Elgammal, "Towards faster and stabilized GAN training for high-fidelity few-shot image synthesis," in *Proc. Int. Conf. Learn. Represent.*, May 2021, pp. 1–14.
- [46] W. S. S. Khine, P. Siritanawan, and K. Kotani, "Generation of compound emotions expressions with emotion generative adversarial networks (EmoGANs)," in *Proc. 59th Annu. Conf. Soc. Instrum. Control Eng. Jpn. (SICE)*, Chiang Mai, Thailand, Sep. 2020, pp. 748–755.
- [47] W. S. S. Khine, P. Siritanawan, and K. Kotani, "Wasserstein based EmoGANs+," in *Proc. Joint 10th Int. Conf. Informat., Electron. Vis., 5th Int. Conf. Imag., Vis. Pattern Recognit.*, Kitakyushu, Japan, Aug. 2021, pp. 1–8.
- [48] S. S. K. Win, P. Siritanawan, and K. Kotani, "Compound facial expressions image generation for complex emotions," *Multimedia Tools Appl.*, vol. 82, no. 8, pp. 11549–11588, Mar. 2023.
- [49] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, Barcelona, Spain, Jan. 2016, pp. 2234–2242.
- [50] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 2818–2826.
- [51] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, Jan. 2017, pp. 6629–6640.
- [52] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Trans. Image Process.*, vol. 21, no. 12, pp. 4695–4708, Dec. 2012.
- [53] W. S. S. Khine, P. Siritanawan, and K. Kotani, "Facial expression features analysis with transfer learning," in *Proc. 14th Int. Conf. Knowl. Syst. Eng. (KSE)*, Nha Trang, Vietnam, Oct. 2022, pp. 1–6.
- [54] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, Jan. 2019, pp. 1–18.
- [55] M. J. Terrill. *For Blacks, Mix of 'Help' Emotions on Oscar Night*. Accessed: Jan. 4, 2024. [Online]. Available: <https://www.deseret.com/2012/2/27/20396135/for-blacks-mix-of-help-emotions-on-oscar-night>
- [56] J. Strauss. *Brendan Fraser Wins Best Actor Oscar in Career Comeback*. Accessed: Jan. 4, 2024. [Online]. Available: <https://apnews.com/article/oscar-2023-best-actor-790bc03b6e935e088c7b6765dd6d2f3f>
- [57] Elsa. *Emma Raducanu: British 18-year-old Makes Tennis History With U.S. Open Final Win*. Accessed: Jan. 4, 2024. [Online]. Available: <https://www.theguardian.com/sport/2021/sep/11/emma-raducanu-makes-tennis-history-with-us-open-final-win>
- [58] G. C. Hin. *Chen Long Emotional As He Announces Badminton Retirement*. Accessed: Jan. 4, 2024. [Online]. Available: <https://www.insidethegames.biz/articles/1137264/chen-long-retires-badminton>
- [59] C. C. Anderson. *The Faces People Make On Roller Coasters Are Hilarious*. Accessed: Jan. 4, 2024. [Online]. Available: <https://www.businessinsider.com/terrified-roller-coaster-faces-photos-2013-4>
- [60] ImUrHuckleBerry. *33 Terrified Roller Coaster Riders That'll Give You a Kick*. Accessed: Jan. 4, 2024. [Online]. Available: <https://www.ebaumsworld.com/pictures/33-terrified-roller-coaster-riders-thatll-give-you-a-kick/85768290/>
- [61] P. Siritanawan, A. Chaiyaroj, P. Tantawanich, K. Sirinaksomboon, and K. Kotani, "Facial expression analysis interpreting emotion in multicultural settings," in *Proc. 62nd Annu. Conf. Soc. Instrum. Control Engineers (SICE)*, Tsu, Japan, Sep. 2023, pp. 1127–1132.
- [62] P. Siritanawan, H. Kojima, and K. Kotani, "Exploring the cultural gaps in facial expression recognition systems by visual features," in *Proc. IEEE Region 10 Conf.*, Chiang Mai, Thailand, Oct. 2023, pp. 1157–1162.
- [63] B. Olimov, S. Karshiev, E. Jang, S. Din, A. Paul, and J. Kim, "Weight initialization based-rectified linear unit activation function to improve the performance of a convolutional neural network model," *Concurrency Comput., Pract. Exper.*, vol. 33, no. 22, p. e6143, Nov. 2021.



WIN SHWE SIN KHINE received the degree from the University of Information Technology (UIT), Myanmar, majoring in software engineering, in 2017, and the M.S. and Ph.D. degrees in information science from Japan Advanced Institute of Science and Technology (JAIST), in 2020 and 2023, respectively. Her research interests include machine learning, computer vision, and face image analysis.



PRARINYA SIRITANAWAN (Member, IEEE) received the B.Eng. and M.Eng. degrees from SIIT, Thammasat University, in 2010 and 2012, respectively, and the Ph.D. degree in information science from Japan Advanced Institute of Science and Technology (JAIST), in 2015. From 2016 to 2018, he was a Postdoctoral Research Fellow with Nanyang Technological University (NTU), Singapore. Prior to his current position, he was a Faculty Member with JAIST, from 2018 to 2024. He is currently with the Graduate School of Science and Technology, Shinshu University, Japan. His research interests include computer vision, machine learning, and robotics. He is a member of the IEEE Society and has served as a committee member and a reviewer on several international conferences and journals in his areas of expertise.



KAZUNORI KOTANI received the B.S. and M.S. degrees in electrical and electronic systems engineering and the Ph.D. degree in information science and control engineering from Nagaoka University of Technology, in 1981, 1983, and 1990, respectively. From 1983 to 1989, he was with the Consumer Products Research Center, Hitachi Ltd. He was a Research Associate with the Department of Electrical Engineering, Nagaoka University of Technology, from 1990 to 1991.

He was a Professor with the School of Information Science, Japan Advanced Institute of Science and Technology (JAIST), in 2024, and he is currently with the Faculty of Transdisciplinary Science, Kanazawa University. His research interests include computer vision, facial image analysis, and CG. He is a member of ITE and J-FACE.

• • •